



Early Life and Obesity

Anthropometric and cardiometabolic risk factors in parents and child obesity in Segamat, Malaysia

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Abstract

Background: There is little evidence regarding risk factors for child obesity in Asian populations, including the role of parental anthropometric and cardiometabolic risk factors. We examined the relation between parental risk factors and child obesity in a Malaysian population.

Methods: We used data from health and demographic surveillance conducted by the South East Asia Community Observatory in Segamat, Malaysia. Analyses included 9207 individuals (4806 children, 2570 mothers and 1831 fathers). Child obesity was defined based on the World Health Organization 2007 reference. We assessed the relation between parental anthropometric (overweight, obesity and central obesity) and cardiometabolic (systolic hypertension, diastolic hypertension and hyperglycaemia) risk factors and child obesity, using mixed effects Poisson regression models with robust standard errors.

Results: We found a high burden of overweight and obesity among children in this population (30% overweight or obese). Children of one or more obese parents had a 2-fold greater risk of being obese compared with children of non-obese parents. Sequential adjustment for parental and child characteristics did not materially affect estimates (fully adjusted relative risk for obesity in both parents: 2.39, 95% confidence interval: 1.82, 3.10, $P < 0.001$; P for trend < 0.001). These associations were not modified by parental or child sex. We found no consistent evidence for associations between parental cardiometabolic risk factors and child obesity.

Conclusions: Parental obesity was strongly associated with child obesity in this population. Further exploration of the behavioural and environmental drivers of these associations may help inform strategies addressing child obesity in Asia.